Cause-Effect Graph based method

(Black Box Testing)

Problem-1

Now, let us try to derive a graph for a problem. The problem is that there are two input values and one output value against each. The first value accepts only character and the character should be either A or B. The second value must be a digit. If the two values has above combination then the output printed is “MESSAGE 1”. If the character in the first value is incorrect, then the output printed is “INCORRECT VALUE 1”. If the character in the second value is not a digit, then the message printed is “INCORRECT VALUE 2”.

Let us try to derive the happy flows of causes and corresponding effect.

Causes are:

C1: Character for first value is A.

C2: Character for first value is B.

C3: Character for second value is a digit.

Effects are:

E1: Output is “MESSAGE 1”. → (C1 OR C2) AND C3.

E2: Output is “INCORRECT VALUE 1” → When first value has neither A nor B.

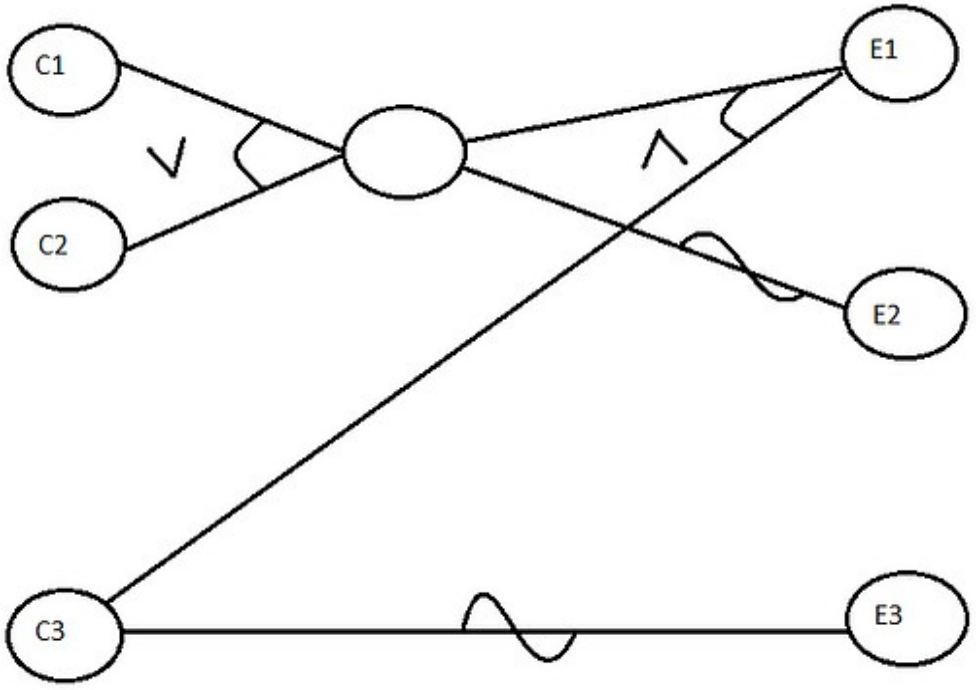
(NOT C1) AND (NOT C2) = NOT (C1 OR C2). Using De-Morgan’s law, NOT of individual causes with AND can be combined as combined NOT of OR of both the causes.

E3: Output is “INCORRECT VALUE 2”. When cause C3 is negated. NOT(C3).

If we try to derive the decision table with both positive and negative values the it would be something like this:

CAUSES	R1	R2	R3	R4	R5	R6
C1: First Value	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
C2: Second Value	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE
C3: Second Digit	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE
EFFECTS						
Output: MESSAGE1	TRUE	TRUE				
Output: INCORRECT VALUE 1			TRUE	TRUE		
Output: INCORRECT VALUE 2					TRUE	TRUE

In the above decision table, each column represents a test case. The same can be derived into a easily understandable graph given below:



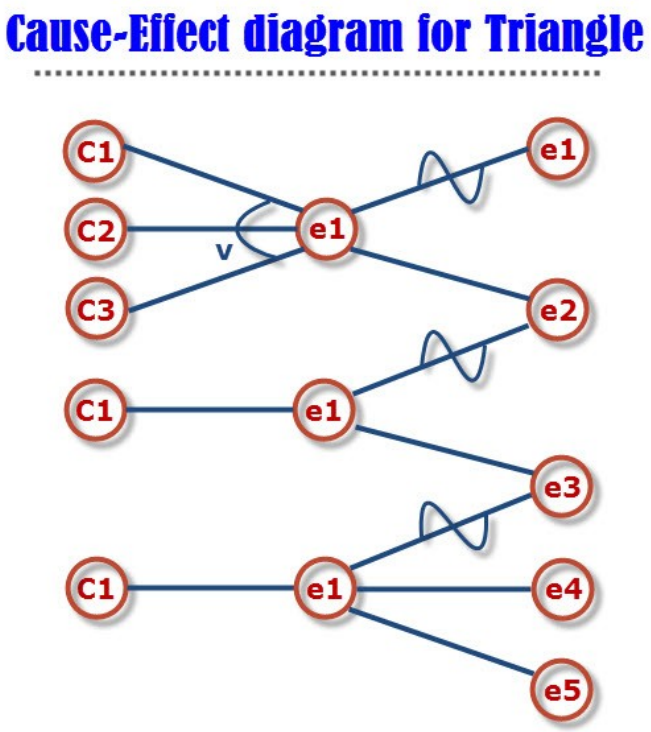
Here, the cause nodes are named and effect nodes are named but the intermediator nod is not required to be named.

We will discuss the constraints in detail in the next blog to understand better. Hope deriving the Cause-Effect graph is clear.

Problem-2

Test cases can be designed for the triangle problem in the following ways

Firstly: Recognize and describe the input conditions (causes) and actions (effect).
The causes allocated by letter "C" are as follows,
C1: Side "x" is less than sum of "y" and "z"
C2: Side "y" is less than sum of "x" and "z"
C3: Side "z" is less then sum of "x" and "y"
C4: Side "x" is equal to side "y"
C5: Side "x" is equal to side "z"
C6: Side "y" is equal to side "z"
The effects designated by letter "e" are as follows,
e1: Not a triangle
e2: Scalene triangle
e3: Isosceles triangle.
e4: Equilateral triangle
e5: Impossible
Secondly: Build up a cause-effect graph



Third: Convert cause-effect graph into a decision table

Conditions	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11
C1: X < Y+Z?	0	1	1	1	1	1	1	1	1	1	1
C2: X < Y+Z?	X	0	1	1	1	1	1	1	1	1	1
C3: X < Y+Z?	X	X	0	1	1	1	1	1	1	1	1
C3: X=Y?	X	X	X	1	1	1	1	0	0	0	0
C4: X=Y?	X	X	X	1	1	0	0	1	1	0	0
C5: X=Y?	X	X	X	1	1	0	0	1	1	0	0
C6: X=Y?	X	X	X	1	0	1	0	1	0	1	0
e1: Not a Triangle	1	1	1								
e2: Scalene											1
e3: IsoScele							1		1	1	
e4: Equilateral				1							
e5: Impossible					1	1		1			

Fourth:11 test cases according to the 11 rules.

Test Case	X	Y	Z	Expected Result
1	4	1	2	Not a triangle
2	1	4	2	Not a triangle
3	1	2	4	Not a triangle
4	5	5	5	Equilateral
5	?	?	?	Impossible
6	?	?	?	Impossible
7	2	2	3	Isosceles
8	?	?	?	Impossible
9	2	3	2	Isosceles
10	3	2	2	Isosceles
11	3	4	5	Scalene